

Flit Expressions, Types, and Equations

CS 350

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Overview

Programming in CS 350

Racket

Flit



THESE ARE YOUR
FATHER'S PARENTHESSES

ELEGANT
WEAPONS

FOR A MORE... CIVILIZED AGE.

Overview

Objectives

- To learn how to use Dr. Racket

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- To learn how to install and use `#lang flit`

Programming in CS 350

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- The Dr. Racket editor

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 - They are different from what we are doing. Beware!

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 - Can evaluate expressions while you're writing your code

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 - You can do a lot with very little

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- Data Types with Variants

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 - NOT always a named function!

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- Languages like C++ and Python use “infix notation” for math

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- In Flit, you write `(f x y z)`
- 99% of what you do in Flit is calling functions

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 - Hierarchical structure: expressions contain expressions contain expressions...
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Square vs. Round vs Curly

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( + 1 2 )  
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 - Use it for the syntax of the new language we're creating

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- JSON is basically just funny-looking parentheses

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(+ 2 7)
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(* 1/3 2/3)
(/ 1 1000000000000000.0)
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```
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9.5
2/9
1e-12
20
1
```

Example

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(> (/ 0 1) 1)
(zero? (- (+ 1 2) (+ 3 0)))
(and (< 1 2) (> 1 0))
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 - *Mutate* value of variable

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- Use functions and recursion to do all computation

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```
if (someCondition)
  doThis
else
  doThat
```

- The if-statement doesn't have a value, but it can mutate (change) the state

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 - e.g. Above, the 2nd if evaluates to 3 because the condition evaluates to true

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 - One works on numbers, one works on expressions

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```
int x;  
x = 3;  
x = 4;
```

- Have $x = 3$ and $x = 4$, so surely $3 = 4$, right?

Equations for IF

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 - The two sides of the equation have same run-time result

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```
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(addOne 10)
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11
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 - Or another expression in brackets

Multi-Argument Functions

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```
(define (isRemainder [x : Number]
                  [y : Number]
                  [remainder : Number])
  : Boolean
  (= remainder (modulo x y)))
(isRemainder 10 3 1)
(isRemainder 10 4 1)
```

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(define (isRemainder [x : Number]
                  [y : Number]
                  [remainder : Number])
  : Boolean
  (= remainder (modulo x y)))
(isRemainder 10 3 1)
(isRemainder 10 4 1)
```

```
#t
```

```
#f
```

Functions (ctd.)

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 - Each argument given as [name : Type]
- Later in the course we'll see another way of defining functions

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Type Inference

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 - Make sure types in your head match types in the code

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Semantics of Function Calls

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Definition (Semantics of Function Calls)

If a function f is defined as:

then for any expression $EXPR$, we have:

where $[x \Rightarrow EXPR]body$ is $body$, with all non-shadowed occurrences of the variable x replaced by $EXPR$

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- Same, as long as there's no side effects

General Syntax

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- Defining a single value has double brackets (`let* ([x 3])`
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 - Kind of awkward but it's what we've got

Definition (Semantics of Let for Single Variables)

For any expression $EXPR$, we have $(\text{let } ([x \ EXPR]) \ \text{body}) = [x \Rightarrow EXPR] \text{body}$ where $[x \Rightarrow EXPR] \text{body}$ is body with all non-shadowed occurrences of x replaced by $EXPR$

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 - We'll see the formal rule in the second part

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- And `letrec`
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- Including referring to themselves
- For now, you only need `let*`